Unit 4: Introduction to Computer Networking

Level: **2**

Unit type: **Internal**

Guided learning hours: **60**

Unit in brief

This unit examines the purpose and features of computer networks, and explores how they are designed, implemented, and tested.

Unit introduction

Computer networking is an important part of our daily business and personal lives that most people take for granted. Without networks, many of the tools and systems used by businesses and individuals would not function: we couldn't surf the web, have video calls, or use a variety of devices such as a smartphone or laptop to communicate with friends and family online, or use games consoles. Security plays an important role in computer networking as organisations need to keep data as secure as possible while protecting their systems against daily threats.

In this unit, you will investigate the network environment and the different security threats it can face, and you will examine the techniques that can be used to defend the network environment and consequently an organisation's business processes.

You will connect computer systems and mobile devices to develop a network environment and will thoroughly test the network to ensure it functions correctly.

Learning aims

In this unit you will:

A Understand a computer network environment and security threats

B Design a computer network

C Implement and test a computer network.

Summary of unit

Learning aim	Key content areas	Assessment approach
A Understand a computer network environment and security threats	**A1** Network environments **A2** Networks in organisations **A3** Security threat types **A4** Computer network security measures	A written report that: • examines network environments in organisations • considers the impact of security threats on organisations • examines measures used to mitigate network security risks.
B Design a computer network	**B1** Factors affecting network design **B2** Hardware components in a network **B3** Software components in a network **B4** Design documentation for a network	A portfolio evidencing: • the design and implementation of a network for a given brief • the testing and optimisation of the network • a rationale for the security measures used.
C Implement and test a computer network	**C1** Implementing a computer network **C2** Testing a computer network	

Content

Learning aim A: Understand a computer network environment and security threats

A1 Network environments

Purpose, types and features of network environments.

- Types of different networks to include Wide Area Network (WAN), Local Area Network (LAN), Personal Area Network (PAN), Virtual Private Network (VPN), Storage Area Network (SAN).
- Physical topologies including star, mesh, tree, ad hoc.
- Methods of connecting a device to a network including: Ethernet, Wi-Fi, Bluetooth®, 3G/4G/5G.
- Features of network environments, e.g. virtualisation, cloud computing, BYOD.
- Protocols and standards including OSI model, TCP/IP, Ethernet standards.
- Hardware components of the network (wired/wireless) including server, routers, switches, hub, cables, nodes, network interface cards (NIC).
- Network software including operating system (client/server), network monitoring, management and troubleshooting tools, application software.

A2 Networks in organisations

Purpose, role and benefits of networks in organisations.

- Communication and accessing information (internal and external to the organisation)
- Supporting business processes.
- Resource sharing and storage/data, e.g. shared printer, cloud computing.
- Collaboration, e.g. online software.
- Entertainment.

A3 Security threat types

Security threats and their impact on organisations.

- Common security threats:
    - internal threat, e.g. unsafe practices, sabotage
    - external threats, e.g. hackers, phishing
    - physical threats, e.g. theft, fire, terrorist action
    - social engineering e.g. phishing (spear phishing/whaling), vishing, smishing, tailgating
    - software driven threats (malware) e.g. Trojan horses, spyware, ransomware, rootkits, worms, keyloggers

- network-based attacks, e.g. distributed denial-of-service (DDoS), denial-of-service (DoS), man in the middle (MitM), DNS Tunneling
- weak cyber security measures, including weak passwords/poor password policy, out of date/no anti-malware sofwtare, poorly configured firewall, latest security patches not applied
- impact of the attacks on organisations including loss of data and information, operational or trading, financial loss, reputation damage.

A4 Computer network security measures

The purpose, function and effectiveness of security tools and techniques that organisations use to mitigate attacks.

- User ID and passwords.
- Encryption/Decryption.
- Software including anti-virus/anti-malware, malware removal tools, firewalls, anti-track.
- Software configuration.
- Access control including files and applications.
- Web/wireless security.
- Patch management.
- Training/risk management.
- IT security policy and procedures.

Learning aim B: Design a computer network

Considerations, methods, and procedures involved when designing a peer-to-peer or client-server network.

B1 Factors affecting network design

- Interpreting user requirements, to include:
    - purpose of the network
    - objectives and activities performed to meet objectives
    - details of the users and their roles/function, including the number of users
    - size and geographical spread/location
    - budget
    - constraints, e.g. costs, hardware and software availability, physical environment (e.g. inside or outside, building layout, materials used in the building).

B2 Hardware components in a network

- Features of hardware components:
    - their role and purpose
    - connection methods
    - communication with other components in the intended computer network.
- Choosing hardware components, to include:
    - computer systems, workstations, servers

- network adapters (wired/wireless)
- router
- hub
- switch
- network cabling, e.g. fibre optics, UTP, STP, coaxial, connectors
- wireless, e.g. Wi-Fi, Bluetooth, NFC, RFID.

B3 Software components in a network

- Features of software components:
  - their role and purpose
  - their relationship with other components in the intended computer network.
- Choosing software components, to include:
  - applications, e.g. internet browsers, email, network utilities (remote management), office applications
  - security, e.g. firewall, anti-virus
  - operating system with appropriate utilities capable of operating in a network environment.

B4 Design documentation for a network

Network requirement considerations and the design and testing documentation process.

- Consideration of different network requirements and preparation of design documentation:
  - computer network set-up and configuration, e.g. IP addressing/subnet masks (if appropriate), network utilities, sharing files and folders, access permissions for users, managing user accounts (add, remove and amend), and sharing hardware/software resources, e.g. internet, printers, storage devices
  - network security (potential threats, how the threats can be mitigated)
  - network user rights, e.g. administration rights, access control rights, shared resource rights
  - network design and testing documentation
  - network diagram, e.g. structure of the network, components, positioning of the components, connection medium and IP addresses (if applicable)
  - test plan with test data to test functionality, e.g. testing the connectivity between all devices on the network.

Learning aim C: Implement and test a computer network

C1 Implementing a computer network

Considerations, methods, and procedures involved when developing a peer-to-peer or client-server network.

- Hardware and software required for implementation:
  - hardware, including:
    - two computer systems/workstations/clients

- a server
- simulated software, such as Packet Tracer
- network adapters
- network cabling
- a hub/router/switch

- software, including:
  - a server operating system
  - operating system, e.g. adding/removing/amending users, sharing files and folders, setting access permissions to files and folders, installing applications, sharing hardware resources, i.e. printer or any other network device
  - security software, e.g. firewall, anti-virus.

C2 Testing the network

Processes for testing computer networks.

- Test computer networks for functionality/connectivity using command tools,

e.g. ipconfig and ping.

- Use utilities/services provided by the operating system to test other aspects of the network, e.g. shared resources, user accounts, access control, security file/folder permissions.
- Obtaining and using client and user feedback, e.g. functionality, usability, requirements, performance.
- Improving the computer network, e.g. performance, capacity, accessibility, scalability, reliability, security.

Assessment criteria

Pass	Merit	Distinction
Learning aim A: Understand a computer network environment and security threats		**A.D1** Evaluate the effectiveness of computer networks to meet organisation requirements, and security measures used to mitigate potential security risks.
A.P1 Identify the role, type and features of different computer networks used in organisations. **A.P2** Identify potential security threats, and security measures used to mitigate risks.	**A.M1** Assess the effectiveness of computer networks to meet organisation requirements, and security measures used to mitigate potential security risks.	
Learning aim B: Design a computer network		
B.P3 Identify the user requirements, hardware and software components for a given brief. **B.P4** Produce a basic design for a computer network from a given brief, to meet user requirements.	**B.M2** Produce a detailed design for a computer network from a given brief, to meet user requirements.	
Learning aim C: Implement and test a computer network		**B.D2** Produce a comprehensive design for a computer network from a given brief, to meet user requirements. **C.D3** Justify approaches taken to connect, configure and apply effective security measures to a network environment in line with own comprehensive plan.
C.P5 Connect, configure and test a basic network environment applying planned basic security measures. **C.P6** Outline approaches and decisions taken in implementing a network environment.	**C.M3** Implement a computer network using own detailed design, testing for functionality and connectivity using feedback from others, including rationale for possible improvements. **C.M4** Explain approaches and decisions taken in implementing a network environment.	

Essential information for assignments

The recommended structure of assessment is shown in the unit summary, along with suitable forms of evidence. *Section 6 Internal assessment* gives information on setting assignments and there is also further information on our website.

There is a suggested maximum number of three summative assignments for this unit. The relationship of the learning aims and criteria is:

Learning aim: A (A.P1, A.P2, A.M1, A.D1)

Learning aims: B and C (B.P3, B.P4, B.M2, B.D2, C.P5, C.P6, C.M3, C.M4, C.D3)

Further information for teachers and assessors

Resource requirements

For this unit, learners must have access to:

- a wide range of hardware and software resources for the practical aspects

of the unit. As a minimum, each learner will need access to computer systems with a network server, hub/router/switch and cabling (if wired), wireless router (if wireless)

- peripherals, e.g. printer/scanner
- facility to connect to the internet
- Learners can also use network simulators, such as Packet Tracer to develop their understanding of the concepts.

Essential information for assessment decisions Learning aim A

For Distinction standard, learners must evaluate the effectiveness of two different

computer networks. They must consider the effectiveness in terms of what an organisation might reasonably expect from such a network.

Learners must consider how different network features are used/applied in each of the identified networks and make a well-reasoned judgement as to their appropriateness. For example, a Personal Area Network (PAN) could be used to connect two offline devices to each other as there would be no need for a server to be implemented.

The connections could be wired to ensure a more secure connection.

Learners must consider the features that have been used by the network. They must make clear judgements as to how well these have been used, considering both positive and negative aspects.

Learners must evaluate the impact potential security threats could have on an organisation, along with how effectively the organisation can mitigate against these risks. For example, an internal threat could damage the reputation of an organisation and in order to mitigate against this, security policies would be required so employees are fully aware of the consequences if they were to do anything that falls out of line of these policies.

At this level learners must make appropriate use of technical vocabulary to effectively support the points they make.

For Merit standard, learners must investigate the effectiveness of the identified networks in meeting the expectations of organisations.

They must make sound judgements as to how well different features of the network help them to meet their intended purpose. For example, a wireless connection to a printer has been used rather than a wired one as it will be in a different room etc. At this level some of the points they make may be more generic and the judgement they make may lack support.

Learners must explore the purpose of the network type and provide a partially supported judgement as to its quality and effectiveness in relation to the intended organisation and what they might reasonably expect from such a network.

Learners must assess the potential security threats and how effectively the organisation could mitigate the risks.

At this level learners must make mostly appropriate use of technical vocabulary to effectively support the points they make in their report.

For Pass standard, learners must explore two different network types and successfully identify the role and features of these within an organisation. For example, an organisation might want to share printing resources so a LAN network would be suitable for this.

Learners must explore security threats that could impact computer networks and how these threats could be reduced. At this level however, these are likely to contain several more generic threats. For example, hackers could be a threat to the network, but a firewall could possibly prevent them from accessing it.

At this level learners will make some appropriate use of technical vocabulary, but this may not be sustained.

Learning aim B

Learners will design, develop, and test their own computer network. They have a choice of designing either a **peer-to-peer** or **client-server** network, depending on hardware and software resource availability.

Learners should be given a brief, which will allow them to design a computer network.

The brief should ideally be written with a 'client' in mind with clearly stated resource requirements and objectives.

The brief must include the following requirements, as a minimum:

- create at least three network users, one of whom must be a network administrator with full control over the computer network
- create at least five shared folders and seven shared files, with different access permissions for different users
- configure network users, e.g. place in groups, set login restrictions, disable accounts
- create at least one shared hardware network resource, e.g. printer, scanner, external storage device
- configure network users, e.g. add more users, remove existing users, change passwords(s), change access rights, set login restrictions, disable user accounts
- setup use of network utilities, e.g. remote desktop, instant messaging, antivirus, firewall.

For Distinction standard, learners will produce a comprehensive set of designs for a computer network in response to a given brief. Their designs must include a requirement brief that provides a comprehensive overview of the network to be

designed, which will include the purpose of the network along with size and possible costs. At this level it is expected that this information will unambiguous and the designs will meet the

user needs. Learners must also consider the hardware and software components to be used in the network.

The design documents must also include set up and configuration, security, user rights, a network diagram and test plan. These documents must be of sufficient clarity and detail that if required a third party could, with minimal difficulty, create the network using the learner's design documentation.

For Merit standard, learners will produce a detailed set of designs for a computer network in response to a given brief. Their designs must include a requirement brief that provides a detailed overview of the network to be designed, which will include the purpose of the network along with size. The designs must meet most of the user needs.

Learners must also consider the hardware and software components to be used in the network.

The design documents must also include set up and configuration, security, user rights, network diagram and test plan. These documents must be of reasonable quality so that if required a third party could, with minimal difficulty, partially create the network using the learner's design documentation.

For Pass standard, learners will produce basic designs for the computer network in response to a given brief. The design documents must also include a requirement brief that provides an overview of the network to be designed, which will include the purpose of the network along with size, but there may be some gaps or omissions. At this level, only some of the user needs may be met. Learners must also consider the hardware and software components to be used in the network.

The design documents may be basic and could include set up and configuration, security, user rights, a network diagram and test plan. These documents may be hampered by lack of clarity or omissions so that that a third party would require assistance when attempting to use them to create the network.

Learning aim C

For Distinction standard, learners must make effective use of hardware and software provided in a network environment to produce a fully functional computer network.

At this level, the network must have sufficient functionality and performance to allow technical audiences such as managers, and/or other IT personnel to provide feedback. Appropriate security measures should have been applied to the network.

The implemented network must fully link to the designs and meet user requirements. A well-reasoned rationale for improvements must be carried out.

The network must be fully tested for functionality and performance. Feedback from others must be obtained and learners must produce evidence to demonstrate they have interpreted the situation correctly, solved any problems, and fully responded to this feedback.

Learners must consider the approaches that have been used when implementing the network. They must make clear, supported judgements as to why these approaches have been taken.

Learners must also consider any possible improvements that could be made to the network; however, these improvements do not have to be implemented. Appropriate technical language will be used effectively throughout.

For Merit standard, learners must make effective use of hardware and software provided in a network environment to produce a functional computer network. At this level, the network must have some functionality and performance to allow technical audiences such as man gers, and/or other IT personnel to provide feedback. Appropriate security measures should have been applied to the network.

The finished network must link to the designs and should meet most user requirements.

The network must be tested for functionality and performance. Feedback from others must be obtained and learners must produce evidence to demonstrate they have interpreted and fully responded to this feedback.

Learners must explore the approaches that have been used when implementing the network. They must make sound judgements as to why these approaches have been taken. Learners must produce a rationale for improvements that could be made to the network, although these improvements do not need to be carried out.

At this level learners must make mostly appropriate use of technical vocabulary to effectively support the points they make in their report.

Learners must also consider any possible improvements that could be made to the network; however, these improvements do not have to be implemented.

For Pass standard, learners must make use of hardware and software provided in a network environment to produce a computer network. At this level, the network may not be fully functional, and performance may be limited but it does address some of the user requirements. Some consideration may have been given to the security measures, but they may not always address the appropriate security threats.

The finished network may not fully link to the designs. The network must be tested for functionality and performance. Feedback from others must be obtained, although it could be brief in parts and learners must produce evidence to demonstrate they have interpreted and responded to this feedback.

Learners must consider the approaches and decisions taken when implementing the network. Some judgements should be made as to why these approaches have been taken but this may not be clear. Learners will generally use appropriate terminology, although this may be inconsistent.

Links to other units and curriculum subjects

This unit links to:

- Unit 3: Setting up a Technology System
- International GCSE/core curriculum in Information Technology.

Opportunities to develop transferable employability skills

In completing this unit, learners will have the opportunity to develop skills in research and planning, problem-solving, responsibility, and communication.